Risk-Aware Machine Learning Under Distribution Shift:
Quantifying Predictive Uncertainty and Selective Abstention in Financial Tabular Streams

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Abstract

Standard supervised learning models rely upon the canonical Independent and Identically Distributed (i.i.d.) assumption, presuming parity between training and deployment environments. In safety-critical tabular domains—such as high-frequency financial transactions, fraud detection, and credit scoring—this premise consistently fails due to covariate shift, concept drift, and subpopulation volatility. High-capacity learners under distribution shift often exhibit overconfidence, making silent yet catastrophic classification errors. This project proposes the **Risk-Aware Selective Classification Pipeline (RASCP)**, a principled framework that integrates post-hoc probability calibration, ensemble epistemic disagreement, and inductive conformal prediction. By equipping automated classifiers with a risk-gated abstention mechanism, RASCP strategically defers ambiguous, out-of-distribution transactions for expert human audit while maintaining automated throughput for safe decisions. Evaluated on the benchmark ULB Credit Card Fraud dataset under four controlled distribution shift regimes, this work establishes empirical bounds on selective classification, disentangles data-level out-of-distribution (OOD) detection from true predictive decision risk, and targets a $\geq 30\%$ reduction in shift-induced monetary loss at an automation rate exceeding 85%.

1 Problem, Motivation and Impact

1.1 The Problem to Address

Supervised machine learning algorithms implicitly assume that training instances $\mathcal{D}_{\text{train}} \sim \mathcal{P}_{\text{train}}(X, Y)$ and future deployment streams $\mathcal{D}_{\text{test}} \sim \mathcal{P}_{\text{test}}(X, Y)$ originate from identical joint probability distributions. In mission-critical financial applications, this assumption is regularly violated by dynamic adversarial forces and macroeconomic shifts:

- **Covariate Shift:** $\mathcal{P}(X)$ diverges while conditional class dynamics $\mathcal{P}(Y \mid X)$ remain invariant (e.g., shifts in merchant categories, customer spending spikes, or seasonal velocity).
- **Concept Drift:** The posterior mapping $\mathcal{P}(Y \mid X)$ mutates dynamically over time, with or without alterations in feature marginals (e.g., fraud syndicates actively altering behavioral signatures to circumvent static thresholds).
- **Subpopulation Shift:** Proportions of underlying customer demographics or transaction channels fluctuate sharply.

When tested under such shifts, modern models—such as Gradient Boosted Decision Trees (GBDTs) and Deep Tabular Neural Networks—suffer from *silent failure modes*. Rather than signaling unreliability, they output uncalibrated, overconfident scores, frequently predicting false negatives or false positives with near-unity certainty. This investigation tackles the fundamental question: **How can a tabular ML system quantify predictive uncertainty under distribution shift and strategically abstain from high-risk decisions without compromising operational efficiency?**

1.2 Importance and Shortcomings of Existing Methods

The ramifications of this failure mode are severe:

1. **Asymmetric Cost Structure:** In transactional fraud, false negatives trigger direct financial theft, chargebacks, and regulatory sanctions. False positives cause customer friction, cart abandonment, and analyst overhead.
2. **Uncalibrated Confidence Scores:** Standard softmax probabilities or decision tree leaf ratios reflect training frequencies rather than true posterior probabilities. Under out-of-distribution (OOD) scenarios, these outputs are notoriously overconfident [2].
3. **Disconnect Between OOD Detection and Decision Risk:** Existing anomaly detection approaches (e.g., Isolation Forest, Mahalanobis Distance) operate purely in unsupervised feature space $\mathcal{P}(X)$. However, an anomalous feature vector may reside in an unambiguous classification region, while an in-distribution sample near the decision boundary may carry catastrophic classification risk.
4. **All-or-Nothing Automation:** Commercial pipelines typically enforce rigid, global score thresholds ($\hat{p} > 0.5$) with no principled fallback to human intervention.

Core Research Hypothesis

Coupling multi-source uncertainty quantification (calibration, ensemble variance, and conformal p-values) with a selective abstention rule yields Pareto-optimal risk-coverage trade-offs, reducing distribution shift error by $\geq 30\%$ while preserving $\geq 85\%$ autonomous throughput.

1.3 Expected Beneficiaries and Impact

- **Financial Risk Analysts:** Receive a disciplined, risk-ordered queue of flagged transactions accompanied by uncertainty scores rather than uncalibrated probabilities.
- **End Consumers:** Protected against automated erroneous card declines and undetected fraudulent exploitation.
- **ML Community & Practitioners:** Gain a reproducible, open-source benchmarking suite for tabular distribution shifts and selective abstention.

2 Dataset Specifications and Controlled Shift Protocols

2.1 Primary Dataset: ULB Credit Card Fraud Benchmark

We employ the authoritative Credit Card Fraud Detection benchmark released by the Machine Learning Group (MLG) at Université Libre de Bruxelles (ULB) and Worldline [1]. The dataset contains real European card transactions recorded over a 48-hour period in September 2013.

Table 1: Primary Dataset Specifications

Attribute	Specification Details
Dataset Name	ULB / Worldline Credit Card Fraud
Modality	Tabular transactional records
Record Count	284,807 transactions
Class Distribution	492 fraud (0.172%), 284,315 legitimate (99.828%)
Dimensionality	30 numerical inputs + 1 binary target label
Features	V_1–V_{28} (PCA projections), `Time`, `Amount`
License	Open Database License (ODbL)
Public Access	Kaggle: `mlg-ulb/creditcardfraud`

2.2 Representative Dataset Sample

Table 2 shows representative records illustrating the severe class imbalance and continuous feature representation.

2.3 Controlled Distribution Shift Generation

To systematically analyze degradation beyond natural temporal drift, the dataset is segmented into training (70%, chronological), validation/calibration (15%), and an in-distribution holdout (15%). The remaining test partition is subjected to four synthetic shift regimes:

Table 2: Representative Normalized Transaction Records

Time (s)	V_1	V_2	V_{28}	Amount	Class (Y)
0.0	−1.3598	−0.0728	−0.0211	€149.62	0 (Legit)
541.0	−2.3122	1.9519	0.1336	€0.00	1 (Fraud)
472.0	−3.0435	3.1573	−0.4359	€529.00	1 (Fraud)
1200.0	1.1918	0.2662	0.0147	€2.69	0 (Legit)

Shift A: Covariate Shift: Injection of parameterized Gaussian noise and affine transformations into sensitive latent principal components (V_1–V_4) and `Amount`:

$$X_{\text{shift}} = X + \mathcal{N}\left(\mu_{\text{shift}}, \sigma^2 \mathbf{\Sigma}_X\right), \quad \sigma \in [0.5, 2.0] \quad (1)$$

Shift B: Subpopulation / Extreme Amount Shift: Partitioning transactions by high transaction values (Amount > 90th percentile), where fraud propensity diverts from micro-transactions.

Shift C: Label Imbalance Perturbation: Artificially compounding fraud incidence from the base 0.172% up to 2.5% to emulate organized fraud campaigns.

Shift D: Temporal Concept Drift: Testing on the terminal 12-hour slice of the 48-hour recording without model updating.

2.4 Ethics, Privacy and Reproducibility

Because features V_1–V_{28} were generated through PCA on confidential financial data, all Personally Identifiable Information (cardholder identities, account numbers, merchants) has been excised. The study operates strictly under ethical standards with fully reproducible open-access data.

3 Proposed Methodology: The RASCP Framework

3.1 Architecture Overview

The proposed **Risk-Aware Selective Classification Pipeline (RASCP)** operates as a sequential three-tiered architecture (Figure 1).

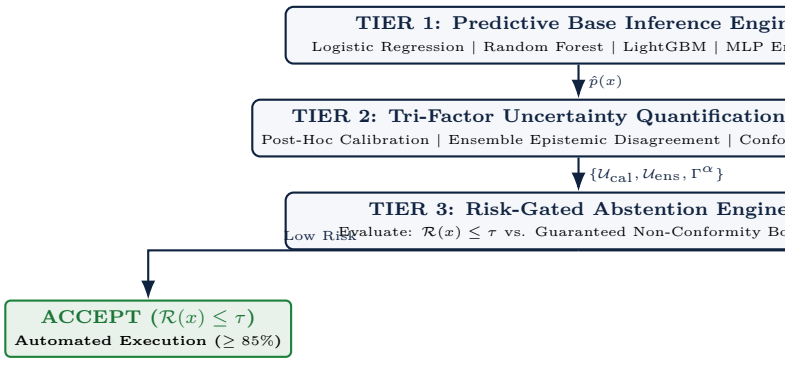


Figure 1: Risk-Aware Selective Classification Pipeline (RASCP) Architecture.

3.2 Base Predictive Classifiers (Tier 1)

To assess model vulnerability across inductive biases, Tier 1 implements:

- **Regularized Logistic Regression (ElasticNet):** Serves as a transparent linear baseline.

- **Random Forest:** Bagged ensemble of decorrelated decision trees producing variance across individual estimators.
- **LightGBM / XGBoost:** Gradient boosted tree ensembles representing state-of-the-art accuracy for tabular classification.
- **Tabular Multi-Layer Perceptron (MLP) & Deep Ensembles:** Neural baseline trained with Monte Carlo Dropout [6].

3.3 Tri-Factor Uncertainty Quantification (Tier 2)

Tier 2 simultaneously estimates aleatoric uncertainty and epistemic ignorance through three complementary mechanisms:

1. **Probability Calibration:** Raw predicted scores $\hat{z}(x)$ are calibrated on held-out validation data via Platt Scaling (logistic sigmoid) and Isotonic Regression [2]:

$$\hat{p}_{\text{cal}}(x) = \sigma(a \cdot \hat{z}(x) + b) \tag{2}$$

2. **Ensemble Epistemic Disagreement:** Measured across M constituent estimators in the ensemble:

$$\mathcal{U}_{\text{ens}}(x) = \frac{1}{M} \sum_{m=1}^{M} \left(\hat{p}_m(x) - \bar{p}(x)\right)^2 \tag{3}$$

   High variance signifies that sub-models disagree due to sparse training support in shifted feature space.
3. **Inductive Conformal Prediction:** Given a desired error rate $\alpha \in (0,1)$, we compute nonconformity scores $s_i = 1 - \hat{p}(x_i, y_i)$ on the calibration set. Conformal prediction sets are defined as:

$$\Gamma^{\alpha}(x) = \{y \in \{0,1\} \mid 1 - \hat{p}(x,y) \leq \hat{q}_{1-\alpha}\} \tag{4}$$

   where $\hat{q}_{1-\alpha}$ is the empirical $(1-\alpha)$-quantile of calibration scores [4]. If $\Gamma^{\alpha}(x) = \{0,1\}$, the model is fundamentally ambiguous, triggering abstention.

3.4 Risk-Gated Abstention Engine (Tier 3)

Let $g(x) \in \{0,1\}$ denote the selection function, where $g(x) = 1$ accepts the prediction and $g(x) = 0$ abstains. We formulate a composite risk score:

$$\mathcal{R}(x) = w_1 \left(1 - \max_y \hat{p}_{\text{cal}}(y \mid x)\right) + w_2 \mathcal{U}_{\text{ens}}(x) + w_3 \mathbb{I}\left(|\Gamma^{\alpha}(x)| > 1\right) \tag{5}$$

A sample is automated if and only if $\mathcal{R}(x) \leq \tau$, where threshold τ is tuned to guarantee a target automation throughput coverage $c = \mathbb{E}[g(X)] \geq 0.85$.

3.5 Key Concrete Contributions

Project Contributions

1. **Controlled Tabular Shift Benchmark:** A turnkey synthesis suite generating four distinct shift modes on financial tabular data.
2. **Empirical Uncertainty Bake-Off:** Systematic head-to-head comparison between Platt scaling, ensemble variance, and conformal prediction in tabular domains.
3. **Pareto Risk-Coverage Optimization:** Derivation of a cost-optimal abstention frontier balancing false negative liability with human analyst bandwidth.
4. **Disentangling OOD from Predictive Risk:** First-principles analysis demonstrating when feature-space anomalies can still be safely automated.

4 Evaluation Protocol and Performance Metrics

4.1 Predictive and Calibration Metrics

Under extreme class imbalance (0.172%), accuracy is uninformative. Models are benchmarked via:

- **AUPRC (Area Under Precision-Recall Curve):** Focuses strictly on the minority fraudulent class.
- **AUROC (Area Under ROC Curve):** Diagnostic power across discrimination thresholds.
- **Expected Calibration Error (ECE):** Partitioning predictions into $K = 15$ confidence bins:

$$\text{ECE} = \sum_{k=1}^{K} \frac{|B_k|}{N} \left|\text{acc}(B_k) - \text{conf}(B_k)\right| \tag{6}$$

- **Brier Score:** Mean squared error between probabilities and one-hot labels.

4.2 Selective Classification and Economic Utility

We evaluate selective classification via:

- **Risk-Coverage Profile & AURC:** Measuring empirical risk $\mathcal{L}(f,g) = \frac{\sum_{i=1}^{N} \ell(f(x_i), y_i) g(x_i)}{\sum_{i=1}^{N} g(x_i)}$ across coverage rates $c \in [0.5, 1.0]$ [3].
- **Cost-Weighted Financial Utility Function:** Quantifying realistic banking losses:

$$\text{Total Cost} = C_{\text{FN}} \cdot \text{FN}_{\text{acc}} + C_{\text{FP}} \cdot \text{FP}_{\text{acc}} + C_{\text{def}} \cdot N_{\text{def}} \tag{7}$$

  where parameterized costs reflect standard banking guidelines: $C_{\text{FN}} = \$100$ (fraud loss), $C_{\text{FP}} = \$5$ (customer friction), and $C_{\text{def}} = \$2$ (human analyst triage).

4.3 Comparative Baselines

1. **Vanilla Classifier:** Standard static threshold $\hat{p} > 0.5$ without abstention.
2. **Unsupervised Feature-Space OOD Detector:** Isolation Forest and Mahalanobis Distance flagging outliers in $\mathcal{P}(X)$.

3. **Random Abstention Baseline:** Abstaining on random transactions at identical coverage fractions c.

4.4 Quantitative Success Criteria

Table 3: Quantitative Success Metrics

Target Metric	Formal Success Criterion
Cost Reduction	$\geq$ 30% lower total financial loss vs. vanilla classifier under Shift A–D.
Automation Rate	Retains $\geq$ 85% coverage ($c \geq 0.85$) without triggering catastrophic error.
Calibration	ECE < 0.05 across all shift test splits post-calibration.
Disentanglement	Proof of non-trivial regions where $\text{OOD}(x) = 1$ but $\mathcal{R}(x) \leq \tau$.

5 Project Roadmap and Timeline

- **Weeks 1–2:** Baseline data ingestion, temporal splitting, and initial classifier benchmark (Logistic Regression, LightGBM).
- **Weeks 3–4:** Implementation of controlled shift injection suite (Covariate, Subpopulation, and Label shift modes).
- **Weeks 5–6:** Integration of Tier 2 uncertainty estimators (Platt scaling, ensemble variance, inductive conformal bounds).
- **Weeks 7–8:** Risk-coverage optimization, financial utility evaluation, error analysis, and final report compilation.

References

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